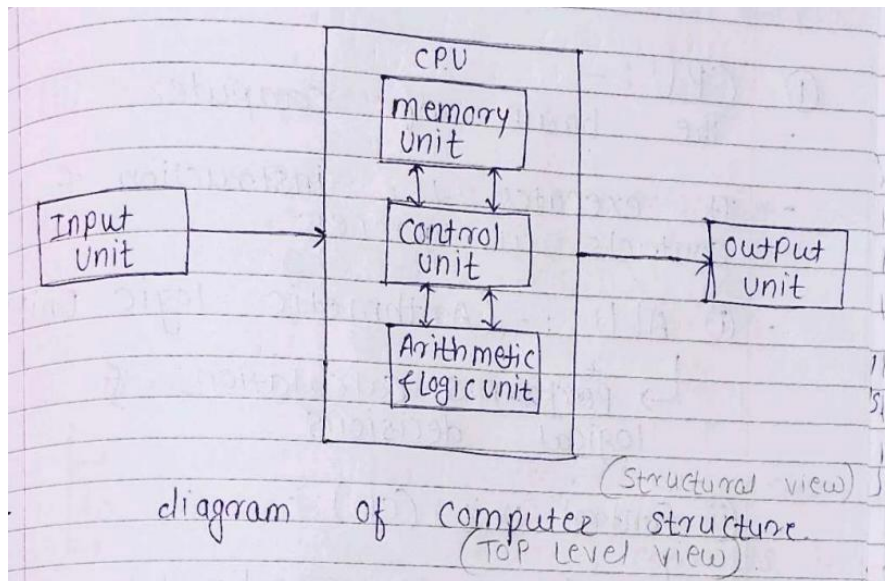


CAO PYQ SOLUTION

Q.1 explain briefly main structure components of computer.

Ans: It is the way in which components are related to each other. It is made up of three main components mainly central processing unit CPU main memory and I/O device



- The components of computer are CPU i o device and memory
- and are connected with interconnection buses
- memory is used to store data
eg. :- semiconductor memory ,magnetic memory, optical memory.
- IO devices are used to take input and give output from CPU
eg. :- Keyboard, scanner, mouse are some input devices and CRT printer etc are output devices

1. CPU :-

- The brain of computer.
- It execute the instruction and control all processes.
- a. ALU (Arithmetic Logic Unit):-
 - performs calculation and logical decisions

b. Control Unit (CU) :-

- direct and coordinates all the computer operations .
- If pages instruction from memory decode them and send ALU, memory ,I/O device.

c. Register :-

- small incredibly fast temporary storage are in CPU .

2. Memory Unit:-

- stores data and instruction

a. Primary memory (main memory RAM)

- temporary holds data and runs program while the computer is on

b. secondary memory :-

- permanent data retention and used for long term like HDD solid state device SSD

3. Input unit :-

- allows user and other system to interact with computer
- Collect raw data , convert it into binary code & process.

Ex.:- keyboard, mouse, scanner.

4. Output Devices:-

- Collect processed binary data and convert results into human language

Ex.:- monitor, printer, speaker.

Q2. Define concept computer organization and computer architecture , structure and function.

Ans: Computer Organization :-

- Refers to the direction or layout ,focus on how the computer does.
- Key elements of computer organisation is control signals interfaces between the computer and peripheral and the specific memory technology Have a hidden attributes and low level hardware details

- It is also known as micro architecture
- It covers Circuit design address catches size and peripheral and implements instruction using physical unit
ex.:- Apple silicon chip design

Computer Architecture :-

- Refers to the conceptual design, structure and behaviour of the computer system as seen by the programmers and focus on what computer does.
- Key elements are: instruction set number of beat used for data representation I/O mechanism and memory addressing technique.
- Have a visible attribute and high level design issue.
- It is also known as instruction set architecture (ISA).
- Covers instruction set data type register and addressing mode and interface between hardware and software

ex.:- X86, ARM or RISC be instruction set

Structure:-

- It is defined at the way in which the individual components of computer are interrelated it described the internal layout and hierarchy of system

Core Components:-

- CPU
- Main memory
- IO unit
- System interconnection

Function :-

- Data processing:- calculating or transforming data .

- Data storage :- keeping data for short term or long term .
- Data movement :-data between system and external environment.
- Control:- managing resources and timing of execution.

Q3. List the addressing modes with example

Ans: 1. Immediate Mode

- Data is given in instruction itself.
- Example: `MOV AX, 2050H`
- Puts 2050H directly into AX.

2. Register Mode

- Data is inside a register.
- Example: `MOV AX, BX`
- Copies data from BX to AX.

3. Direct Mode

- Instruction gives memory address.
- Example: `MOV AX, [5000H]`
- Takes data from address 5000H.

4. Register Indirect Mode

- Register holds the memory address.
- Example: `MOV AX, [BX]`
- If BX=3000H, takes data from 3000H.

5. Indexed Mode

- Address = Index register + number.
- Example: `MOV AX, [SI + 10H]`
- Adds 10H to SI for address.

6. Implied Mode

- No address needed in instruction.
- Operation works on fixed location.
- Example: `STC` sets Carry Flag directly.

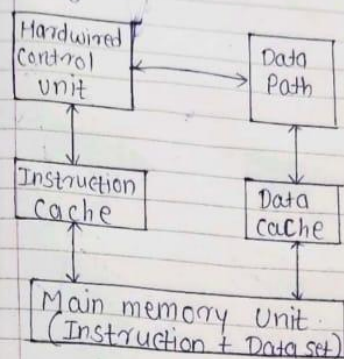
Q4. Differentiate between RISC & CISC

Ans:

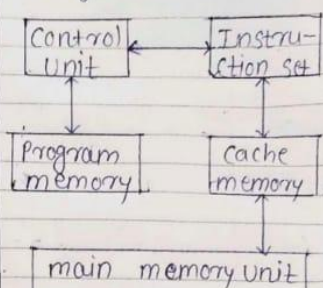
Q. Difference B/w RISC & CISC.

RISC	CISC
① Reduced Instruction Set Computer	① Complex Instruction Set Computer
② Fixed size Instruction	② variable size Instructions
③ executes in one clock cycle	③ executes in multiple clock cycle.
④ highly optimized software	④ highly complex hardware.
⑤ many registers	⑤ few register
⑥ Less addressing modes	⑥ many addressing modes
⑦ easy pipeline	⑦ difficult Pipeline
⑧ many lines of code	⑧ few lines of code
⑨ complex compiler	⑨ simple compiler
⑩ consumes very less power	⑩ consumes high power
⑪ example :- phones, tablets.	⑪ de example :- desktop laptop.

⑬ diagram :-



⑬ diagram:-



Q5. Calculate 1's and 2's complement of following a)12 b)18 c)27 d)9

Ans:

(a) 12.

- Binary of 12 is 00001100
- 1's complement means invert the bit 0 to 1 & 1 to 0.
i.e. 1's of 12 is 11110011
- 2's complement is 1's + 1
→ 11110011 + 1
= 11110100

(b) 18

- Binary of 18 is 00010010
- 1's complement = 11101101
- 2's complement = 1's + 1
= 11101101 + 1
= 11101110

(c) 27

- Binary of 27 is 00011011
- 1's complement = 11100100
- 2's complement = 1's + 1
= 11100100 + 1
= 11100101

(d) 9.

- Binary for 9 is = 00001001
- 1's complement is = 11110110
- 2's complement is = 1's + 1
= 11110110 + 1
= 11110111

Q6. Write steps of binary restoring algorithm with example

Ans: Binary division restoring algorithm is used to divide two unsigned binary numbers

steps

step 1) start :- set $A=0$

M = divisor

Q = dividend

count = total bits

step 2) shift left :- shift the combined register pair $[A,Q]$ one bit to the left

step 3) subtract:- $A \leftarrow A - M$

step 4) check sign of A :- if A is negative put zero in Q then restore A by adding M back and if A is positive put 1 in Q

step 5) Loop:- reduce count by 1 and repeat. Do this until count become zero

step 6) final result:- The quotient is stored in Q and reminder in A

Example: Divide 8 by 3 ($n = 4$)			
- Dividend (Q): 1000 (8) - Divisor (M): 0011 (3) $-M$: 1101 (2's complement of 3)			
Iteration	Step	Accumulator (A)	Dividend (Q)
Initial	Start	0000	1000
1	Shift Left [A, Q]	0001	000_
	$A = A - M$	1110 (Negative)	
	Restore A , $Q_0 = 0$	0001	0000
2	Shift Left [A, Q]	0010	000_
	$A = A - M$	1111 (Negative)	
	Restore A , $Q_0 = 0$	0010	0000
3	Shift Left [A, Q]	0100	000_
	$A = A - M$	0001 (Positive)	
	Set $Q_0 = 1$	0001	0001
4	Shift Left [A, Q]	0010	001_
	$A = A - M$	1111 (Negative)	
	Restore A , $Q_0 = 0$	0010	0010
Final Results:			
- Quotient (Q): $0010_2 = 2$ - Remainder (A): $0010_2 = 2$			

Q7. Explain cache memory with property “locality of reference” & replacement algorithm.

Ans:

1. Cache Memory

Cache memory is a small, high-speed buffer located between the CPU and main memory

(RAM). Its primary purpose is to store frequently accessed data and instructions to reduce the average time required to access memory.

- Locality of Reference

The efficiency of cache memory relies on the principle of Locality of Reference, which

suggests that programs tend to reuse data and instructions they have used recently. It has two main types:

a. Temporal Locality

- Means "same thing again".
- If we use some data now, we will use it again soon.

Example: A loop runs same instructions many times.

b. Spatial Locality

- Means "nearby things".
- If we use one data, we will use data near it also.

Example: Reading array elements one by one, like A, A, A.[0][1][2]

2. Cache Replacement Algorithms

Since cache memory is much smaller than main memory, it eventually becomes full. When new data needs to be loaded into a full cache, a replacement algorithm decides which existing block to evict.

Common algorithms include:

- Least Recently Used (LRU): This is the most popular algorithm. It evicts the block that hasn't been accessed for the longest period of time, assuming it is the least likely to be needed again soon.
- First-in, First-Out (FIFO): This replaces the oldest block in the cache (the one that was loaded first), regardless of how often it has been accessed.

- Least Frequently Used (LFU): This replaces the block that has been accessed the fewest number of times over a period.
- Random Replacement: A block is chosen at random for eviction. While simple, it doesn't utilize locality as effectively as LRU.

Q8. What is mapping function ? explain direct mapping method with neat diagram

Ans: Mapping Function

- A mapping function is the algorithm or method used to transform a main memory address into a specific location within the cache memory.
- Since the cache is much smaller than main memory, mapping functions define how multiple memory blocks share the limited number of cache lines.

Direct Mapping Method

In Direct Mapping, each block of main memory is mapped to exactly one specific line in the cache. If a block is stored in the cache, it must occupy that specific line.

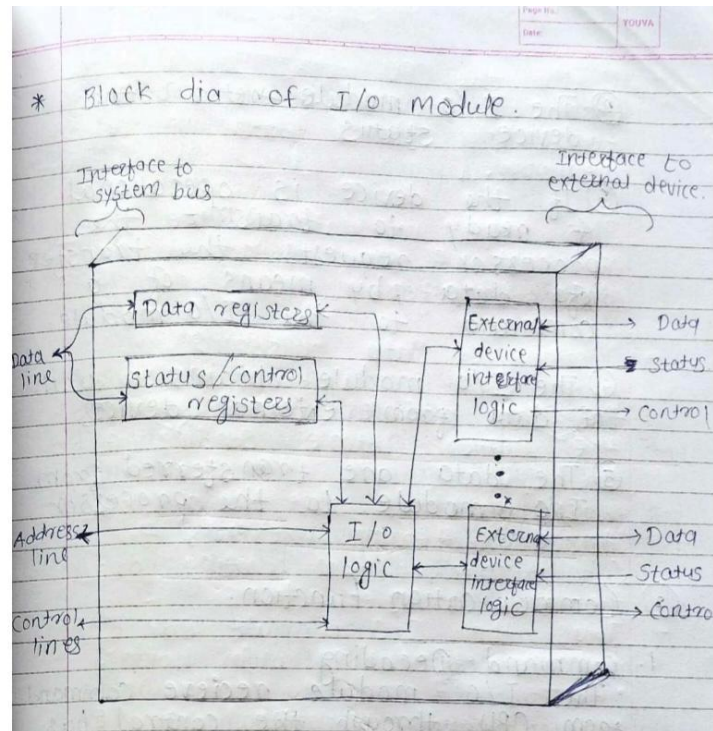
- Logic: The mapping is determined by the formula:

$$i = j(\text{mod } m)$$

Where:

- i = Cache line number
- j = Main memory block number
- m = Total number of lines in the cache

Purpose: It hides the complex details of the peripheral device from the CPU.

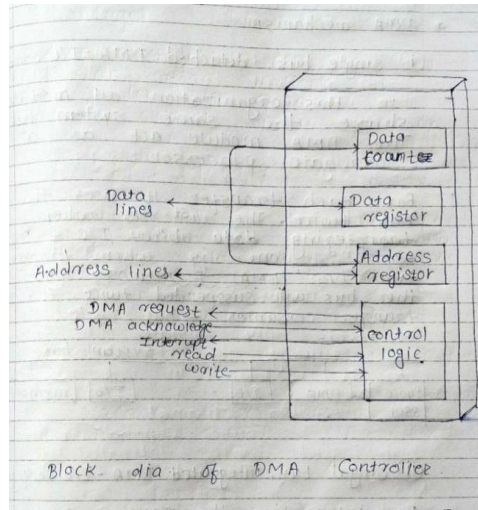


2. Direct Memory Access (DMA)

DMA is a technique that allows an I/O module to transfer data directly to or from main memory without continuous involvement from the CPU.

Mechanism: The CPU "delegates" the transfer to a DMA controller by providing the starting address and the number of words to be transferred. Once the transfer is complete, the controller sends an interrupt to the CPU.

Benefit: It significantly improves system performance by freeing up the CPU to perform other tasks while data is being moved.



3. I/O Channels

An I/O Channel is an advanced version of an I/O module. It is essentially a specialized processor with its own instruction set designed specifically for managing I/O operations.

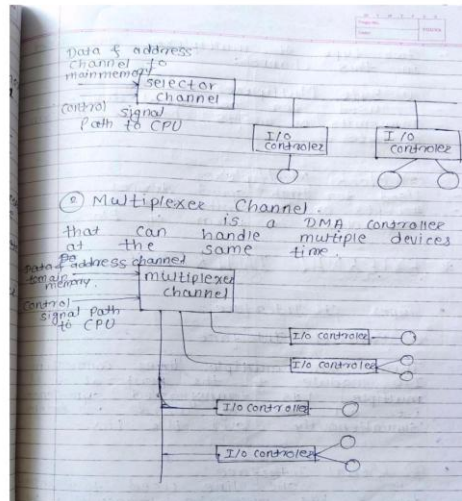
Role: It can execute an entire "I/O program" stored in main memory. The CPU only needs to issue a single command to start the sequence.

Types: Selector Channel: Manages high-speed devices one at a time.

Multiplexor Channel: Manages multiple slow-speed devices simultaneously

.

Types:-
 ① selector channel
 Controls multiple high speed devices dedicated to transfer data with one of device.
 - Each device is handled I/O controller or I/O module.



Q10. List the phase of microinstruction execution

Ans: microinstruction:-

Micro instruction is small instruction in control memory that generate control signal for one CPU cycle

1. Fetch Phase

In this phase, the Microprogram Counter (MPC) provides the address of the next microinstruction to the Control Store. The microinstruction is then retrieved and loaded into the Microinstruction Register (MIR).

2. Execution Phase

This phase is responsible for translating the bits of the microinstruction into actions. It is generally divided into two simultaneous tasks:

- **Control Signal Generation:** The control bits in the MIR are decoded to produce signals that operate the CPU's data path (ALU, registers, and memory gates).
- **Next Address Generation (Sequencing):** The control unit determines the address of the next microinstruction based on the current

instruction's address field and status flags (like Zero or Carry) from the ALU.

- Execution step

1. fetching micro instruction
2. Decode micro operation
3. Execute micro operation
4. Determine next micro instruction address
5. Repeat until instruction execution complete

- Advantages

Simple design .

Easy modification of instruction set .

Supports complex instruction.

Q11. State the basic fundamental units of computer and give brief explanation with diagram.

Ans: (refer the answer of **Q1.**)

Q12. Short note on 1) PC 2)IR 3)MAR 4)MDR

Ans:

i) Program Counter (PC)

- Definition: A register that holds the memory address of the next instruction to be executed.
- Function: Once an instruction is fetched, the PC is automatically incremented to point to the next sequential instruction, ensuring the program flows in the correct order.

ii) Instruction Register (IR)

- Definition: A register that holds the actual instruction currently being decoded or executed.
- Function: After the CPU fetches an instruction from memory (using the address in the PC), it places that instruction into the IR. The control unit then examines the IR to determine what action to take.

iii) Memory Address Register (MAR)

- Definition: A register that holds the memory location (address) of data that needs to be accessed.
- Function: Whether the CPU needs to read an instruction or write data to RAM, it first loads the specific memory address into the MAR. This register is directly connected to the address bus.

iv) Memory Data Register (MDR)

- Definition: A register that holds the actual data being transferred to or from memory.
- Function: It acts as a buffer. When reading from memory, the data found at the MAR's address is placed into the MDR. When writing, the CPU places data in the MDR before sending it to memory. It is connected to the data bus.

Q13. Explain in detail instruction fetching and execution.

Ans(refer CAO Unit 1 pdf page no. 19 - 23)

Q14. Explain type of operand and operation with suitable instruction of assembly language

Ans: In assembly language, an instruction typically consists of an Opcode (the operation to perform) and Operands (the data or locations being acted upon).

1. Types of Operands

Operands indicate where the data for an instruction is located.

- Immediate Operand: The data is a constant value specified directly within the instruction.

- Example: MOV AX, 05H (Moves the hex value 05 into register AX).

- Register Operand: The data is stored in one of the CPU's internal registers.

- Example: MOV AX, BX (Copies the content of register BX into AX).

- Memory Operand: The data is located in the system RAM. The address can be specified directly or indirectly via a register.

- Example: MOV AX, [1000H] (Moves data from memory address 1000H into AX).

2. Types of Operations

Operations define the action the CPU performs on the operands.

- Data Transfer Operations: Used to move data between registers, memory, and I/O ports without altering the value.

- Example: MOV, PUSH, POP, IN, OUT.

- Instruction: MOV CX, DX

- Arithmetic Operations: Perform mathematical calculations like addition, subtraction, multiplication, or division.

- Example: ADD ,SUB ,MUL ,DIV ,INC ,DEC .

- Instruction: ADD AX,02H (Adds 2 to the value in AX).

- Logical Operations: Perform bitwise logic or shifts.

- Example: AND,OR ,XOR ,NOT ,SHL (Shift Left), SHR (Shift Right).

- Instruction: (Performs bitwise AND between AL and BL).

- Control Transfer (Branching) Operations: Alter the sequential flow of execution based on conditions or loops.

- Example: JMP (Unconditional), JZ (Jump if Zero),CALL ,RET .

- Instruction: JZ Lable (Jump to 'Label' if the Zero Flag is set).

Q15. Explain RISC & CISC architectures

Ans: (refer CAO Unit 2 notes pdf page no 20 - 24)

Q16. write short note on 1) magnetic disk 2)magnetic tape 3) optical memory .

Ans: 1) Magnetic Disk

A magnetic disk is a storage device used to store data permanently. It is made of a circular disk coated with magnetic material. Data is stored in the form of magnetic patterns on the disk surface.

Features:

- Provides direct access to data.
- Stores large amounts of information.
- Faster than magnetic tape.
- Commonly used in computers as hard disks.

Advantages:

- High storage capacity.
- Fast data retrieval.
- Reliable for long-term storage.

Examples: Hard Disk Drive (HDD), Floppy Disk.

2) Magnetic Tape

A magnetic tape is a storage medium made of a plastic strip coated with magnetic material. Data is recorded sequentially on the tape.

Features:

- Stores large amounts of data at low cost.
- Data is accessed in sequence.
- Mainly used for backup and archival purposes.

Advantages:

- Very economical.
- Suitable for long-term data storage.
- Large storage capacity.

Disadvantages:

- Slow data access.
- Cannot directly access data.

Examples: Backup tapes used in banks and data centers.

3) Optical Memory

Optical memory is a storage device that uses laser light to read and write data. Data is stored in the form of tiny pits and lands on the disk surface.

Features:

- Portable and easy to use.
- Data is read using a laser beam.
- Resistant to magnetic fields.

Advantages:

- Lightweight and durable.
- Good for storing multimedia files.
- Easy to carry and share.

Examples: CD (Compact Disc), DVD (Digital Versatile Disc), Blu-ray Disc.

Q17. Explain RAM & DRAM with neat diagram.

Ans:

RAM (Random Access Memory)

Definition:

RAM is the main memory of a computer. It stores data and programs that are currently being used by the CPU. It is a **temporary memory**, so data is lost when the power is turned off.

Features:

- Fast memory.
- Data can be accessed directly from any location.
- Used while running programs.

- Volatile memory (data is lost when power is off).

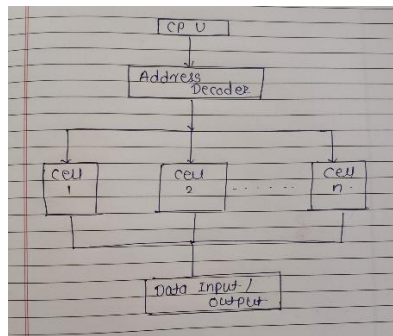
Advantages:

- High speed.
- Improves computer performance.
- Easy access to data.

Examples: RAM used in laptops, desktops, and smartphones.

Conclusion:

RAM helps the computer work quickly by storing the data and instructions needed at the current time.



DRAM (Dynamic Random Access Memory) – 6 Marks

Definition:

DRAM is a type of RAM that stores data using tiny capacitors. The stored data must be refreshed continuously to prevent it from being lost.

Features:

- Dynamic memory.
- Requires regular refreshing.
- Higher storage capacity than SRAM.
- Less expensive than SRAM.

Advantages:

- Low cost.
- Large memory capacity.

- Widely used as the main memory in computers.

Disadvantages:

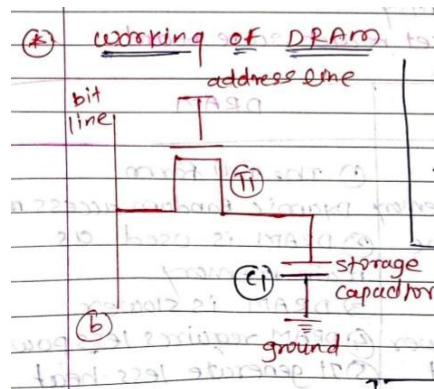
- Slower than SRAM.
- Needs continuous refreshing.

Applications:

Used as the main memory in computers, laptops, and mobile devices.

Conclusion:

DRAM is a low-cost and high-capacity type of RAM used in most computers for primary memory storage.



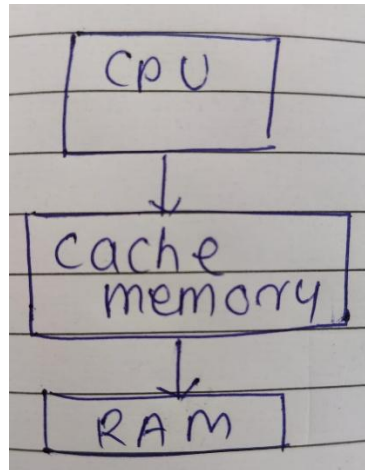
Q18. Explain cache memory, set associative mapping with neat diagram.

Ans: 1) Cache Memory (6 Marks)

Definition

Cache Memory is a very small and very fast memory located between the CPU and Main Memory (RAM). It stores frequently used data and instructions so that the CPU can access them quickly.

Block Diagram



Working

- CPU first checks data in Cache Memory.
- If data is found, it is accessed quickly.
- If data is not found, it is fetched from RAM.

Advantages

- Very high speed.
- Reduces CPU waiting time.
- Improves overall computer performance.

Conclusion

Cache memory acts as a fast buffer between CPU and RAM, making the computer work faster.

2) Set Associative Mapping (6 Marks)

Definition

Set Associative Mapping is a cache mapping technique in which cache memory is divided into several sets. A block from main memory can be placed in any location within a particular set.

Working

- Cache is divided into sets.

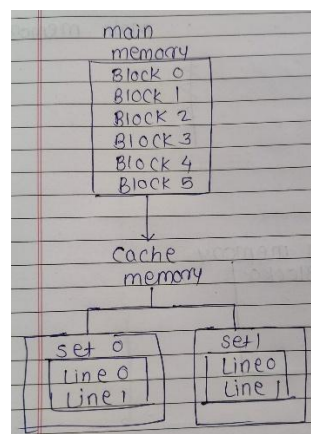
- Each memory block maps to one specific set.
- Within that set, the block can be stored in any line.

Advantages

- Faster than direct mapping.
- Fewer cache conflicts.
- Better utilization of cache memory.

Example

If cache has 2 sets, Block 0, 2, 4 can go to Set 0 and Block 1, 3, 5 can go to Set 1.



Conclusion

Set Associative Mapping combines the advantages of Direct Mapping and Associative Mapping, providing good speed and efficient cache usage.

One-Line Exam Answer

Cache Memory: High-speed memory between CPU and RAM.

Set Associative Mapping: A cache mapping method where a memory block can be stored in any line of a particular set.

Q19. Explain Micro programmed control unit .

Ans: **Microprogrammed Control Unit**

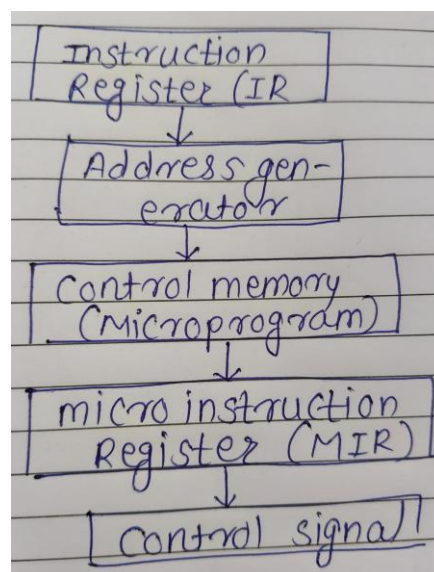
Definition

A microprogrammed control unit is a method of designing a CPU's control unit where control signals are generated by reading pre-programmed microinstructions from a dedicated, internal memory called Control Memory. Instead of using complex digital logic circuits, it uses a software-like, programmatic approach.

In other words a Microprogrammed Control Unit is a part of the CPU that controls all operations of the computer using a set of small instructions called microinstructions stored in a control memory.

It generates control signals step by step to execute machine instructions.

Neat Block Diagram



Working

1. The instruction is stored in the **Instruction Register (IR)**.
2. The **Address Generator** selects the address of the required microinstruction.
3. The microinstruction is fetched from **Control Memory**.
4. It is loaded into the **Microinstruction Register (MIR)**.

5. The control signals generated by the microinstruction control the CPU operations.
-

Advantages

- Easy to design and modify.
 - Complex instructions can be implemented easily.
 - Less hardware is required.
-

Disadvantages

- Slower than hardwired control unit.
 - Requires extra memory for storing microprograms.
-

Q20. Write short note on I/O channel and direct memory access (DMA).

Ans: (refer Q9. ANSWER) {for more details refer CAO Unit 5 notes pdf}

Q21. Explain microinstruction sequencing

Ans:

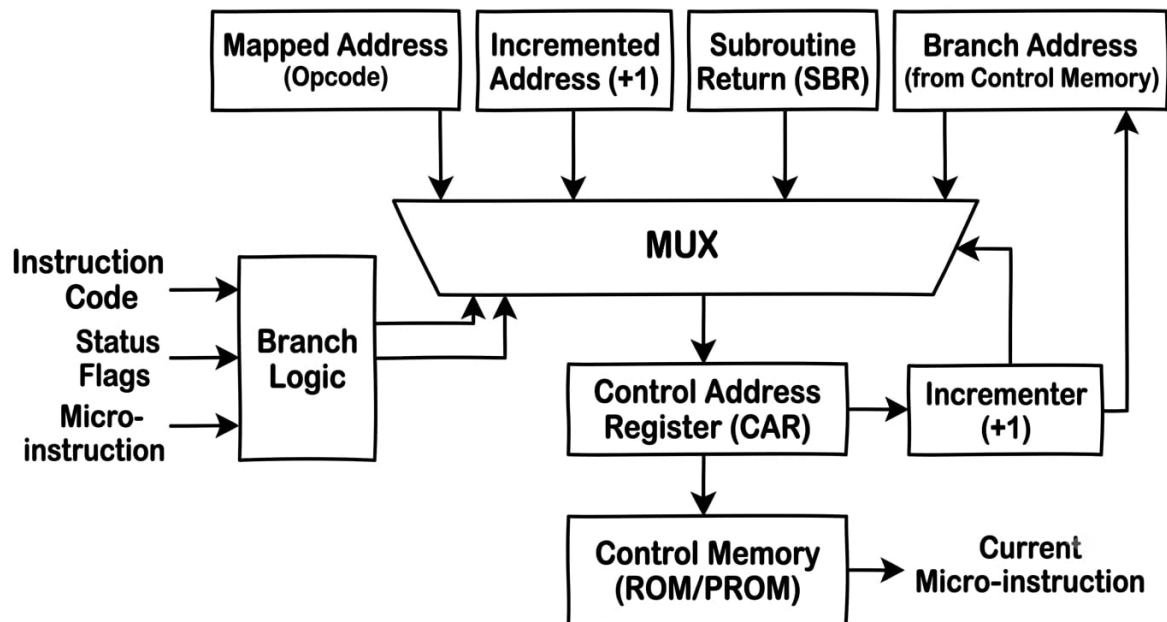
Microinstruction Sequencing

Definition

Microinstruction Sequencing is the process of deciding which microinstruction should be executed next in a microprogrammed control unit.

It controls the order in which microinstructions are fetched and executed.

Block diagram



Working

1. A microinstruction is fetched from the control memory.
2. It is executed and generates control signals.
3. The sequencing logic determines the address of the next microinstruction.
4. The next microinstruction is fetched and executed.
5. This process continues until the machine instruction is completed.

Functions of Microinstruction Sequencing

- Determines the next microinstruction address.
- Maintains the correct execution sequence.
- Supports branching and looping in microprograms.
- Ensures proper execution of machine instructions.

Advantages

- Flexible control unit design.
 - Easy to modify microprograms.
 - Supports complex instructions.
-

Q22. Discuss the role of buses in computer organization. Explain the different types of buses and their function.

Ans: (Refer CAO Unit 1 notes pdf page no. 11 - 13)

Q23. Describe the fetch-decode execution cycle in the CPU. Explain the role of each stage in the execution of instruction cycle.

Ans: (refer CAO Unit 1 notes pdf page no. 19 - 23)

Q24. Discuss the assembly language programming, how does it differ from the high-level programming language?

Ans: (Refer CAO Unit 2 notes pdf page no. 7 - 10)

Q25. Describe the concept of addressing mode, in the list of addressing mode.

Ans: (Refer CAO Unit 2 notes pdf page no. 11 - 18)

Q26. Explain different types of memories with examples.

Ans: **Memory:-**

Memory is a storage area in a computer used to store data, instructions, and results.

1) Primary Memory (Main Memory)

Primary memory is directly accessed by the CPU. It stores data and programs currently in use.

Types:

a) RAM (Random Access Memory)

- Temporary memory.
- Data is lost when power is OFF.

Example: 8 GB RAM in a laptop.

b) ROM (Read Only Memory)

- Permanent memory.
- Stores startup instructions.

Example: BIOS stored in ROM.

2) Cache Memory

- Very fast memory located between CPU and RAM.
- Stores frequently used data.

Example: L1, L2, and L3 Cache.

Advantage: Increases processing speed.

3) Secondary Memory

- Used for permanent storage.
- Large storage capacity.

Examples:

- Hard Disk (HDD)

- Solid State Drive (SSD)
 - Pen Drive
 - Memory Card
-

4) Magnetic Memory

- Stores data using magnetic material.

Examples:

- Hard Disk
- Magnetic Tape

Use: Data storage and backup.

5) Optical Memory

- Uses laser light to read and write data.

Examples:

- CD
- DVD
- Blu-ray Disc

Use: Storing music, videos, and software.

Q27. Explain different types of instruction set.

Ans: **Instruction set :-**

The complete collection of instruction that are understood by CPU are called as instruction sets

Here is a detailed breakdown of the four main categories of computer instructions:

1. Data Processing

- These are the "thinking" instructions. They perform the actual calculations and logical decisions.
- **Arithmetic:** Basic math like addition, subtraction, multiplication, and division.
- **Logic:** Comparisons like "AND," "OR," and "NOT." This allows the computer to decide if something is true or false.
- **Where it happens:** These occur inside the ALU (Arithmetic Logic Unit) of the CPU.

2. Data Storage (Main Memory)

The CPU has very little space to keep information. These instructions move data between the CPU and the Main Memory (RAM).

- **Load:** Bringing data from the memory into a CPU register so it can be worked on.
- **Store:** Taking a finished result from the CPU and saving it back into the memory for later use.
- **Importance:** This ensures the computer can handle large amounts of data that don't fit inside the processor itself.

3. Data Movement (I/O)

A computer is useless if it can't talk to the outside world. These instructions handle communication with Input/Output devices.

- **Input:** Taking data from a keyboard, mouse, or sensor.
- **Output:** Sending data to a monitor, printer, or speaker.

- **Example:** When you type a letter, an I/O instruction moves that character from the keyboard into the system.

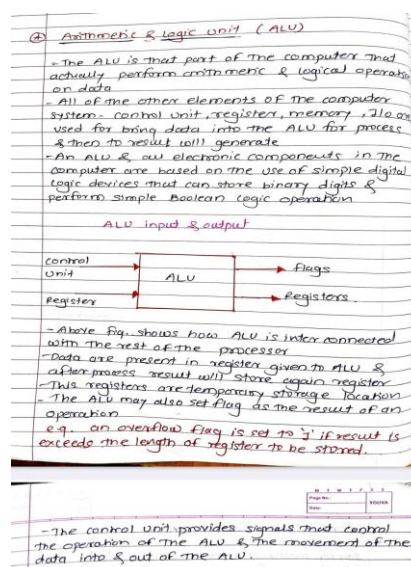
4. Program Flow Control

Normally, a computer executes instructions one by one in a straight line. Flow control instructions change that "path."

- **Branches/Jumps:** Telling the computer to "skip to step 10" or "go back to step 1."
- **Conditions:** "If the result is zero, jump to the end of the program."
- **Loops:** Repeating a set of instructions multiple times until a task is finished.
- **Significance:** This gives the computer its flexibility, allowing it to run complex software rather than just acting like a simple calculator.

Q28. Explain the function of ALU.

Ans: (Refer CAO Unit 3 notes pdf page no. 1)



Q29. Briefly explain the following representations: Sign magnitude, Two's complement, One's complement.

Ans:

1) Sign Magnitude Representation

Definition

In **Sign Magnitude**, the leftmost bit (MSB) represents the sign:

- **0 = Positive**
- **1 = Negative**

The remaining bits represent the magnitude (value).

Example (8-bit)

+5=00000101

-5 = 10000101

Advantages

- Easy to understand.

Disadvantages

- Two representations of zero (+0 and -0).

2) One's Complement Representation

Definition

In **One's Complement**, a negative number is obtained by changing all 0s to 1s and all 1s to 0s.

Advantages

- Simple method for representing negative numbers.

Disadvantages

- Has two zeros:
 - $+0 = 00000000$
 - $-0 = 11111111$
-

3) Two's Complement Representation

Definition

In **Two's Complement**, a negative number is obtained by:

1. Finding the One's Complement.
2. Adding 1 to it.

Advantages

- Only one zero representation.
- Easy arithmetic operations.
- Most commonly used in computers.

Disadvantages

- Slightly more complex than Sign Magnitude.

Comparison Table

Method	Positive 5	Negative 5
Sign Magnitude	00000101	10000101
One's Complement	00000101	11111010
Two's Complement	00000101	11111011

Q30. What are the four essential elements of a number in floating point notation?

Ans:

A **floating-point number** is used to represent very large or very small numbers in a computer.

A floating-point number has **four essential elements**:

1) Sign of the Number

- Indicates whether the number is positive or negative.
- Represented by + or −.

Example: +3.14, −3.14

2) Mantissa (Significand)

- Contains the significant digits of the number.
- Represents the actual value of the number.

Example:

In 3.14×10^2 , the mantissa is **3.14**.

3) Sign of the Exponent

- Indicates whether the exponent is positive or negative.
- Represented by + or −.

Example:

In 3.14×10^{-2} , the exponent sign is −.

4) Exponent

- Specifies the position of the decimal point.
- Determines the magnitude of the number.

Example:

In 3.14×10^2 , the exponent is 2.

Q31. Explain the types of semiconductor memory technologies.

Ans: **Semiconductor memory:-**

Semiconductor memory is a memory made using semiconductor electronic circuits. It is used to store data and instructions in computers.

1) RAM (Random Access Memory)

- RAM is a **temporary memory**.
- Data is lost when the power is turned OFF.
- Used to store programs and data currently being used.

Types of RAM

a) SRAM (Static RAM)

- Stores data using flip-flops.
- Faster than DRAM.
- Expensive and consumes more space.

Example: Cache Memory.

b) DRAM (Dynamic RAM)

- Stores data using capacitors.
- Needs continuous refreshing.
- Slower but cheaper than SRAM.

Example: Main Memory (RAM) in computers.

2) ROM (Read Only Memory)

- ROM is a **permanent memory**.
- Data remains even when power is OFF.
- Stores startup instructions.

Types of ROM

a) PROM (Programmable ROM)

- Can be programmed only once.

b) EPROM (Erasable Programmable ROM)

- Data can be erased using ultraviolet (UV) light and reprogrammed.

c) EEPROM (Electrically Erasable Programmable ROM)

- Data can be erased and rewritten electrically.

d) Flash Memory

- Advanced type of EEPROM.
- Faster and widely used.

Examples: Pen Drive, SSD, Memory Card.

Advantages

- Fast access speed.
- Reliable and compact.
- Low power consumption.

Disadvantages

- High cost.
- Data loss when power is OFF (RAM).
- Limited storage capacity compared to secondary storage devices.

Q32. Explain memory hierarchy in computer system.

Q33. Explain the working of optical memory.

Ans: **Optical Memory:-**

Optical Memory is a storage device that uses **laser light** to read and write data on an optical disk.

Examples: CD, DVD, Blu-ray Disc.

Working of Optical Memory

1) Data Storage

- Data is stored on the surface of the optical disk in the form of **pits** (small holes) and **lands** (flat areas).

2) Laser Beam Reading

- A laser beam is focused on the rotating disk.
- The laser light falls on pits and lands.

3) Reflection of Light

- Lands reflect more light.
- Pits reflect less light.

4) Data Detection

- A sensor detects the reflected light.
- Different reflections are converted into binary data (0s and 1s).

5) Data Processing

- The binary data is sent to the computer.
- The computer converts it into usable information such as text, audio, or video.

Advantages

1. Large storage capacity.
2. Easy to carry and use.
3. Not affected by magnetic fields.

Disadvantages

1. Slower than hard disks and SSDs.
2. Can be damaged by scratches.
3. Limited rewrite cycles (for some disks).

Q34. Explain the programmed I/O model and interrupt-driven I/O.

Ans: (Refer CAO Unit 5 notes pdf page no. 10 - 13)

Q35. Explain input/output organization of computer system.

Ans: (Refer CAO Unit 5 notes pdf page no. 1 - 3)

Q36. Is RISC better than CISC ? Illustrate your answer with example of processor.

Ans: (Refer CAO Unit 2 notes pdf page no. 20 - 24)

Q37. Elaborate the concept of SRAM and DRAM, memory with typical memory cell structure.

Ans: 1) SRAM (Static Random Access Memory)

Definition

SRAM is a type of RAM that stores data using **flip-flops**. It does not need refreshing as long as power is supplied.

Features

- Very fast memory.
- Does not require refreshing.
- More expensive.
- Used in cache memory.

Advantages

- High speed.
- Low access time.

Disadvantages

- Costly.
 - Requires more space.
-

2) DRAM (Dynamic Random Access Memory)

Definition

DRAM is a type of RAM that stores data using a **capacitor and transistor**. It needs continuous refreshing to retain data.

Features

- Requires periodic refreshing.
- Slower than SRAM.
- Less expensive.

- Used as main memory in computers.

Advantages

- Low cost.
- High storage capacity.

Disadvantages

- Slower than SRAM.
- Needs refreshing.

SR.no.	SRAM	DRAM
1.	Made of 6 Transistors	Made of 1 Transistor + 1 Capacitor
2.	Stores data as voltage states	Stores data as an electrical charge
3.	Does not require continuous refreshing.	Requires constant refreshing thousands of times per second.
4.	It is faster	It is slower
5.	Used in cache memory	Used in main memory
6.	Requires more power	Requires less power
7.	Expensive	Cheap
8.	Require minimum time to access data	Requires less time to access data
9.	Generates less heat	Generates more heat
10.	Non-destructive read	destructive read

Q38.Difference between the hardwired and microprogrammed implementation of control unit.

Ans: (Refer CAO Unit 5 notes pdf page no.24 [bottom part])

Q39. Explain the concept of instruction pipelining

Ans:

Q40. Differentiate between interrupt-driven I/O and programmed I/O and DMA.

Ans:

No.	Programmed I/O	Interrupt-Driven I/O	DMA (Direct Memory Access)
1	CPU controls all I/O operations.	CPU starts I/O and waits for interrupt.	DMA controller manages data transfer.
2	CPU continuously checks device status.	Device sends interrupt when ready.	Device transfers data directly to memory.
3	CPU is fully busy during transfer.	CPU can do other work until interrupt occurs.	CPU is free during most of the transfer.
4	Slow method.	Faster than Programmed I/O.	Fastest method.
5	High CPU involvement.	Medium CPU involvement.	Very low CPU involvement.
6	Wastes CPU time.	Better CPU utilization.	Best CPU utilization.
7	No interrupt required.	Interrupt signal is required.	DMA controller is required.
8	Suitable for small data transfer.	Suitable for moderate data transfer.	Suitable for large data transfer.
9	Simple to implement.	More complex than Programmed I/O.	Most complex method.
10	Low system performance.	Better system performance.	Highest system performance.

